

Literature Status for arXiv:2202.01745

Run fa_banach_001, lane 2

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Status

Already answered in the literature. This packet records that the model questions surfaced by O’Loughlin’s 2022/2023 paper on symmetric matrix representations of truncated Toeplitz operators are answered by a separate later paper by the same author, arXiv:2607.14019.

Source Question

The source paper is Ryan O’Loughlin, *Symmetric matrix representations of truncated Toeplitz operators on finite dimensional spaces*, arXiv:2202.01745, later published in *Linear Algebra and its Applications* 666 (2023), 11–28 [1].

In its introduction, the source recalls the finite-dimensional model question:

Is every symmetric matrix unitarily equivalent to a direct sum of TTOs?

After giving a negative answer to a modified-Clark-basis formulation, the source poses the refined Section 3 conjecture labelled **newconj**: if a complex symmetric matrix M is unitarily equivalent to a truncated Toeplitz operator, does there exist an inner function u and a C_θ -real basis for K_u such that M is the matrix representation of a TTO with respect to that basis?

Supporting Answer

The supporting paper is Ryan O’Loughlin, *Semialgebraic Dimension and Truncated Toeplitz Models for Complex Symmetric Matrices*, arXiv:2607.14019, submitted 2026-07-15 [2]. Its abstract and introduction state that it resolves the finite-dimensional model question negatively. More precisely, Corollary 4.2 proves that, for every $n \geq 10$, not every $n \times n$ symmetric matrix is unitarily equivalent to a direct sum of truncated Toeplitz operators.

The same supporting paper also treats the refined representation question. In Section 5 it restates the source’s refined conjecture as Conjecture 5.1 and then proves the following theorem: if B is a finite Blaschke product, A is a TTO on K_B , and $M \in \mathcal{CS}_n$ is unitarily equivalent to A , then there exists a C_B -real orthonormal basis $v_1, \dots, v_n$ of K_B such that $M = [A]_{v_1, \dots, v_n}$.

Scope

The broad direct-sum model question is answered negatively in dimensions $n \geq 10$. The refined single-TTO representation question is answered positively in finite dimensions when the unitary equivalence to a TTO is already known to exist. This is a literature-status packet only; it should not be counted as an original proof, partial result, or counterexample produced by the run.

References

- [1] R. O’Loughlin, *Symmetric matrix representations of truncated Toeplitz operators on finite dimensional spaces*, Linear Algebra Appl. 666 (2023), 11–28; arXiv:2202.01745.
- [2] R. O’Loughlin, *Semialgebraic Dimension and Truncated Toeplitz Models for Complex Symmetric Matrices*, arXiv:2607.14019, submitted 2026-07-15.